

CUSTOMER SEGMENTATION ANALYSIS

Project Report

1. Project Overview

Customer Segmentation Analysis is a data analytics project using RFM (Recency, Frequency, Monetary) analysis to understand customer purchasing behaviour and identify different customer segments.

2. Objectives

- Analyze customer purchasing behaviour.
- Segment customers using RFM analysis.
- Identify valuable and at-risk customers.
- Analyze customer revenue and count by segment.
- Create an interactive Power BI dashboard.

3. Technologies Used

Python: Data analysis and processing

Pandas: Data cleaning and transformation

Jupyter Notebook: RFM analysis workflow

Power BI: Dashboard and visualization

CSV: Customer segmentation data

4. Methodology

The project follows an RFM-based workflow. Customer transaction data is prepared and analyzed to calculate Recency, Frequency and Monetary values. RFM results are then used to assign customers to meaningful business segments and the final dataset is visualized in Power BI.

5. Customer Segments

- **Champions:** Highly valuable and engaged customers.
- **At Risk:** Customers who may need re-engagement.
- **Dormant / Lost:** Customers with weak recent engagement.
- **Price-Sensitive / Low-Value:** Customers with relatively lower value.

6. Power BI Dashboard

The Power BI dashboard presents the RFM results through KPI cards and customer-segment visualizations. The completed dashboard shows approximately **1.2K Total Customers**, **19.92M Total Revenue**, and **16.60K Average Order Value**.

Dashboard Visuals

- Total Customers
- Total Revenue
- Average Order Value
- Customer Distribution by Segment
- Customer Count by Segment
- Revenue by Customer Segment
- Recency vs. Monetary Value by Customer Segment

7. Key Insights

The dashboard provides a clear comparison of customer value and purchasing behaviour across the four segments. Champions contribute strongly to customer value and revenue, while At Risk and Dormant / Lost groups provide opportunities for customer re-engagement. The segment-level charts also help compare customer count with revenue contribution.

8. Business Use

The segmentation can support targeted customer strategies. Champions can be focused on for retention and loyalty, At Risk customers can receive re-engagement campaigns, Dormant / Lost customers can be targeted with win-back activities, and Price-Sensitive / Low-Value customers can receive relevant offers.

9. Conclusion

This project demonstrates how RFM analysis and Power BI can be combined to understand customer behaviour and support data-driven customer segmentation. The final dashboard provides an easy-to-understand view of customer distribution, revenue and purchasing behaviour.

10. Project Deliverables

- Customer_Segmentation_Analysis.ipynb – Data analysis and RFM segmentation
- rfm_final_segments.csv – Final customer segmentation dataset
- Customer Segmentation Analysis.png – Power BI dashboard image
- README.md – Project documentation